

AVIK DE, PH.D.

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PROFESSIONAL SUMMARY -----

Robotics executive and hands-on technologist with deep expertise across legged mobility, embedded real-time systems, and full-stack robot architecture. Co-founded and scaled a commercial robotics company from 2 to 70 people while leading core platform architecture spanning mechanics, electronics, firmware, and high-performance control software. Background combines academic research in dynamic locomotion with production deployment of fielded robotic systems.

TECHNICAL SKILLS -----

Management & Leadership. Grew startup from 2-70 people; managed up to 10 direct reports; agile methodology; product roadmaps; cross-functional engineering lead; KPIs; vendor relationships.

Software. Architecture and implementation of Ghost Robotics robot and mobility software and SDK; high-performance concurrent C++ running on embedded platforms for real-time control; hardware drivers; RT Linux; ONNX; TensorRT; GStreamer; physics simulation with Bullet physics.

Firmware. Architecture and implementation of multi-process real-time control systems; STM32; comms using EtherCAT, Ethernet, USB, RS485, USART, SPI, I2C; RT Linux; advanced motor control (FOC, field-weakening, etc.).

Electrical. Architecture; PCB schematic & layout (Eagle PCB, Altium Designer); high-power electronics including brushless motor controllers; MCU-equipped robot mainboards.

Mechanical. Design architecture; motor and drivetrain selection using analysis & simulations; Solidworks.

PROFESSIONAL EXPERIENCE -----

Adjunct Associate Professor of Electrical & Systems Engineering | UPenn 2015 — Present

- Research in robotics and reinforcement learning
- Guest lecturer in graduate-level reinforcement learning course

Co-founder & Chief Science Officer | Ghost Robotics 2015 — Present

- Assembled and **managed** an engineering organization from founding team to 70 people.
- **Technical owner** of robot platform **architecture**: drivetrain, actuation, electronics, and computing subsystems
- **Leadership** in establishing innovation and commercial applications divisions.
- Mobility, perception, and estimation **algorithms**; proven customer-renowned rough-terrain stability.
- **Hands-on** implementation of low-level software, electronics, firmware.
- Directed transition from research prototypes to robust commercial systems.

Postdoctoral Fellow | Harvard University Microrobotics Lab (Dr. Rob Wood) 2019 — 2020

- Research on optimization-based RoboBee design and hierarchical flapping flight control.

Ph.D. Researcher | UPenn KodLab (Dr. Daniel Koditschek) 2010 — 2017

- Research on modular and bio-inspired control of hopping and running.
- Developed new lab mechatronics infrastructure, Penn Jerboa and Delta Hopper robots.
- Work led to spin-out of Ghost Robotics, a successful startup.

Undergraduate Researcher | JHU LIMBS Lab (Dr. Noah Cowan) 2007 — 2010

- Research on neuromechanical control of juggling, SLAM on graphs, leading to 3 publications.

EDUCATION -----

Ph.D. University of Pennsylvania, Philadelphia, PA. Advisor: Dr. Daniel E. Koditschek 2017

M.S. (Mech Engg), B.S. (Engg Mech), B.S. (Appl Math). Johns Hopkins Univ, Baltimore, MD. 2010

HONORS & AWARDS -----

- [Philadelphia Business Journal 40 Under 40](#) -- 2026
- [Entrepreneur Of The Year Greater Philadelphia](#) -- 2025
- Outstanding reviewer awards: [RAL](#), [Bioinspiration & Biomimetics](#), [IOP publishing](#)
- Best Doctoral Dissertation Award for the 2017-2018 Academic Year. UPenn Electrical and Systems Engineering dept -- 2018
- [James S. McDonnell Postdoctoral Fellowship](#) Winner. \$200k award for 2-3 years -- 2017

- Robert George Gerstmeier Award. For academic achievement (JHU Mech. Engg. Department) 2008-09
- Provost's Undergraduate Research Award. For summer research (JHU) -- 2008-09
- Tau Beta Pi, Pi Tau Sigma Engg. honor societies. Inductee (JHU) -- 2008
- International Scholarship. Merit-based full-tuition scholarship (JHU) -- 2006-10

SELECTED PUBLICATIONS -----

(For a full list of publications and citation metrics, see [Google Scholar](#).)

- De, A., Stewart-Height, A., & Koditschek, D. E., *Task-Based Control and Design of a BLDC Actuator for Robotics*, **IEEE Robotics and Automation Letters (RA-L)** — design and control of brushless DC actuators for robotic applications.
- De, A. & Koditschek, D. E., *Vertical Hopper Compositions for Preflexive and Feedback-Stabilized Quadrupedal Bounding, Pacing, Pronking, and Trotting*, **International Journal of Robotics Research (IJRR)** — hybrid dynamical systems for robust legged gaits.
- Kenneally, G., De, A., & Koditschek, D. E., *Design Principles for a Family of Direct-Drive Legged Robots*, **IEEE Robotics and Automation Letters (RA-L)** — methodology for high-performance direct-drive legged platforms.
- Burden, S. A., De, A., & Koditschek, D. E., *A Hybrid Dynamical Extension of Averaging and Its Application to the Analysis of Legged Gait Stability*, **International Journal of Robotics Research (IJRR)** — extending averaging methods for hybrid robotic locomotion stability.
- Vasilopoulos, V., Arslan, O., De, A., & Koditschek, D. E., *Sensor-Based Legged Robot Homing Using Range-Only Target Localization*, **IEEE/ROBIO Conference** — reactive navigation for legged robots in unknown environments.
- De, A., McGill, R. and Wood, R.J., 2022. *An efficient, modular controller for flapping flight composing model-based and model-free components*. **International Journal of Robotics Research (IJRR)** — hierarchical Model-predictive control of RoboBee flapping flight
- Topping, T. T., Vasilopoulos, V., De, A., & Koditschek, D. E., *Mode-Reactive Template-Based Control in Planar Legged Robots*, **Journal Article** — hierarchical control strategies for legged locomotion.

SELECTED INVITED TALKS -----

Invited speaker at major robotics venues including ICRA, IROS, RSS workshops, CMU RI Seminar, IFRR Quadruped Colloquium, and academic/industry seminars on legged robot design, power-efficient locomotion, shared autonomy, actuator limits, and bio-inspired control architectures (2015–2024).

TEACHING, MENTORING & SERVICE -----

- Instructor (UPenn/Coursera Robotics Specialization – Mobility & Capstone)
- Guest Lecturer, UPenn graduate robotics and dynamical systems courses
- Mentor to undergraduate, M.S., and Ph.D. researchers (ICRA/IROS publications)
- Reviewer: IJRR, T-RO, RA-L, Automatica, Bioinspiration & Biomimetics, and major robotics conferences
- Outreach: FIRST Lego League judge; animal welfare volunteer

SELECTED OPEN-SOURCE PROJECTS -----

- **Robotics teaching and demonstration.** [Hierarchical manipulation pipeline](#)
- **Computer architecture libraries.** [TinyXPU](#) modular systolic array
- **STM32 microcontrollers.** Main developer for [koduino](#) Arduino core for STM32 microcontrollers, which was adopted by the stm32duino community before the first-part HAL libraries became stable; [stm32ld](#) fork supporting auto-reset.
- **Upstream contributions.** Minor contributor to [bullet3](#), [nanopb](#), [osqp](#), [qhull](#), [TinyTorch](#).